Stochastic Fast Krasnosel'skiĭ–Mann Iteration with Unbiased Oracle Evaluations

Working draft — October 15, 2026

Abstract

We consider the problem of finding a fixed point of a nonexpansive operator in a real Hilbert space using conditionally unbiased stochastic oracle evaluations. The operator and its fixed-point set are deterministic, whereas the oracle responses and the iterates are random. We focus on a stochastic implementation of the Fast Krasnosel'skiĭ–Mann method that uses one new oracle evaluation per iteration and retains the preceding evaluation. The objective is to identify noise conditions under which the last-iterate fixed-point residual decays at an accelerated almost-sure rate and the iterates converge weakly almost surely. This initial draft specifies the problem, the oracle model, and the iteration; stochastic convergence theorems and sharp noise thresholds remain to be established.

Key words. nonexpansive operator, fixed-point problem, Fast KM, stochastic oracle, almost-sure convergence.

1 Introduction

We study the problem of finding a fixed point of a nonexpansive operator $T : \mathcal{H} \to \mathcal{H}$ using noisy evaluations of T. Such problems arise from convex optimization and operator splitting, where the fixed points of a suitable update map encode solutions of the underlying problem [1, 4]. The map T is fixed throughout the iteration. Randomness enters through the computation of its values. For instance, in smooth convex stochastic optimization, a sampled gradient provides an unbiased estimate of the gradient of a deterministic expectation objective [2, Section 4.3].

1.1 Background and motivation

The Krasnosel'skiĭ–Mann (KM) iteration is a basic method for nonexpansive fixed-point problems. Boţ and Nguyen [1, Algorithm 2.1 and Theorems 3.4–3.5] proposed the Fast KM method, which combines inertial and operator-difference terms. Under their parameter conditions, the deterministic method satisfies $\|x_k - Tx_k\| = o(1/k)$ and its iterates converge weakly to a fixed point. These guarantees motivate the study of the same iteration with stochastic oracle evaluations.

Replacing exact evaluations by noisy responses introduces a specific dependence on past errors. In the implementation considered here, the previous oracle response is stored and used again in the next update. Consequently, the update contains both the current error and the preceding error, with opposite signs and different coefficients. The preceding error is already known before the current query, so conditional unbiasedness does not remove all error terms from a one-step energy estimate. The question is how this dependence affects the noise conditions needed for almost-sure residual rates and convergence of the iterates.

1.2 Related work

Bravo and Cominetti [2] analyze stochastic KM iterations in finite-dimensional normed spaces, including almost-sure convergence and bounds on the expected fixed-point residual. Yang et al. [4, Theorem 1] study stochastic inertial KM with one oracle response per update. Their parameter schedule depends on a prescribed horizon, and their main result controls the root-mean-square residual at the last iterate. Their oracle model allows a predictable bias; we restrict attention to its conditionally unbiased case. Almost-sure accelerated

residual rates are also available for other inexact fixed-point methods, including the AFP framework of Nguyen-Trung and Tran-Dinh [3, Theorem 2], under the assumptions of that theorem. Thus, an almost-sure accelerated rate alone does not distinguish the present research question.

Our focus is the stochastic Fast KM recurrence specified in Section 3, run indefinitely with parameters independent of a terminal horizon. We seek noise conditions for the deterministic residual rate to persist almost surely, together with weak convergence of the iterates. Comparing those conditions across noise distributions and query implementations is a further objective. No sharp threshold or novelty claim is asserted in this draft.

2 Problem formulation and stochastic oracle

2.1 Nonexpansive fixed-point problem

Let $\mathcal{H}$ be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$.

Assumption 2.1 (Operator and solution set)**.** The deterministic operator $T : \mathcal{H} \to \mathcal{H}$ is nonexpansive:

$$\|Tx - Ty\| \le \|x - y\|, \qquad x, y \in \mathcal{H}. \tag{2.1}$$

Its fixed-point set $\operatorname{Fix} T := \{x \in \mathcal{H} : Tx = x\}$ is nonempty.

The problem is

$$\text{find } x^\star \in \mathcal{H} \quad \text{such that} \quad Tx^\star = x^\star. \tag{2.2}$$

We measure accuracy by the fixed-point residual

$$r(x) := \|x - Tx\|, \qquad r_k := r(x_k). \tag{2.3}$$

For $V := \mathrm{Id} - T$, problem (2.2) is equivalently $Vx^\star = 0$. Nonexpansiveness of T is equivalent to $1/2$-cocoercivity of this V:

$$\langle Vx - Vy, x - y \rangle \ge \tfrac{1}{2}\|Vx - Vy\|^2, \qquad x, y \in \mathcal{H}. \tag{2.4}$$

Indeed, expanding $\|(x-y)-(Vx-Vy)\|^2 \le \|x-y\|^2$ gives (2.4), and the expansion is reversible. The residual does not by itself provide a bound on the distance to $\operatorname{Fix} T$ without an additional error-bound condition.

2.2 Conditionally unbiased oracle

Work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a filtration $(\mathcal{F}_k)_{k \ge 0}$. The sigma-algebra $\mathcal{F}_k$ contains all information available immediately before query k. At an $\mathcal{F}_k$-measurable point x_k, the oracle returns

$$\widehat{T}_k(x_k) = Tx_k + e_k. \tag{2.5}$$

All Hilbert-space-valued random variables are strongly measurable. The error e_k is square-integrable and $\mathcal{F}_{k+1}$-measurable. We impose conditional unbiasedness:

$$\mathbb{E}[e_k \mid \mathcal{F}_k] = 0 \qquad \text{almost surely.} \tag{2.6}$$

This is the specialization $b_k = 0$ of the oracle model in [4, Section 2.2, equation (6)]. Independence of errors across iterations is not assumed.

Let $(\sigma_k^2)_{k \ge 0}$ be a deterministic sequence of finite, nonnegative numbers satisfying

$$\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] \le \sigma_k^2 \qquad \text{almost surely.} \tag{2.7}$$

We distinguish two noise conditions used in the research plan:

(N2) Uniformly bounded conditional variance: $\sup_{k \ge 0} \sigma_k^2 < \infty$.

(N6) Weighted square summability, specialized here to fixed-step Fast KM:

$$\sum_{k=0}^{\infty}(k+1)^2\sigma_k^2<\infty. \tag{2.8}$$

The shift by one includes the initialization error and does not change the asymptotic summability requirement. We use (N6) as a candidate sufficient condition to investigate, not as an established convergence theorem for the recurrence below. A bounded variance budget in (N2) alone does not require the errors to vanish.

Remark 2.2 (The target is deterministic). The operator T and the set $\operatorname{Fix} T$ are deterministic. The oracle responses and the iterates are random. A limit selected by the iteration may consequently be a random variable with values in $\operatorname{Fix} T$. The model does not require the sampled mappings to have a common fixed point or to be individually nonexpansive.

2.3 A stochastic optimization example

Let $f(x)=\mathbb{E}_\zeta[F(x,\zeta)]$ be convex and differentiable, with L-Lipschitz gradient, $L>0$, and a nonempty set of minimizers. For $0<\gamma\le 2/L$, the operator

$$Tx=x-\gamma\nabla f(x) \tag{2.9}$$

is nonexpansive and $\operatorname{Fix} T=\operatorname{argmin} f$. Assume that a sampled gradient g_k at x_k satisfies $\mathbb{E}[g_k\mid\mathcal{F}_k]=\nabla f(x_k)$ and has finite conditional second moment. This unbiasedness must be justified by the sampling model and, when using $\nabla_x F$, the interchange of differentiation and expectation. Then

$$\widehat{T}_k(x_k)=x_k-\gamma g_k,\qquad e_k=-\gamma\big(g_k-\nabla f(x_k)\big) \tag{2.10}$$

satisfies (2.6). This is the stochastic-gradient interpretation of KM described in [2, Section 4.3].

Remark 2.3 (Unbiasedness of the full operator evaluation). An unbiased gradient passed through a nonlinear proximal map need not produce an unbiased estimate of the resulting fixed-point operator. For example, let $P(u)=\max\{u,0\}$ and let ξ be an integrable, nonzero symmetric scalar noise. Then

$$\mathbb{E}[P(-\gamma\xi)]=\frac{\gamma}{2}\mathbb{E}|\xi|>0=P(0).$$

Thus, for a proximal-gradient map, condition (2.6) must be checked for the complete oracle response. The existence of a deterministic splitting formulation alone does not verify this condition.

3 Fast KM iteration

3.1 The Krasnosel'skiĭ–Mann iteration

For the fixed-point problem (2.2), the Krasnosel'skiĭ–Mann (KM) iteration takes the form

$$x_{k+1}=(1-\lambda_k)x_k+\lambda_k Tx_k,\qquad k\ge 0, \tag{3.1}$$

where $x_0\in\mathcal{H}$ and $\lambda_k\in(0,1)$ is a relaxation parameter. Each update is a convex combination of the current iterate and its image under T. Under Assumption 2.1, for a fixed relaxation parameter $\lambda_k\equiv\lambda\in(0,1)$, the iterates converge weakly to a point in $\operatorname{Fix} T$ and the fixed-point residual satisfies $\|x_k-Tx_k\|=o(k^{-1/2})$; see [1, Section 1, equation (1.3)].

3.2 The Fast KM method

Boţ and Nguyen [1] accelerate the KM iteration using Nesterov's momentum updates. For fixed parameters $\alpha>2$ and $0<s\le 1$, and initial points $x_0,x_1\in\mathcal{H}$, their Fast KM method is

$$\begin{aligned}x_{k+1}=&\left(1-\frac{s\alpha}{2(k+\alpha)}\right)x_k+\frac{(1-s)k}{k+\alpha}(x_k-x_{k-1})+\frac{s\alpha}{2(k+\alpha)}Tx_k\\&+\frac{sk}{k+\alpha}(Tx_k-Tx_{k-1}),\qquad k\ge 1.\end{aligned} \tag{3.2}$$

This is equation (2.7) in [1], specialized to a nonexpansive operator. The update involves both the difference of successive iterates and the difference of successive operator values. Under Assumption 2.1, [1, Theorems 3.4–3.5] give

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \rightharpoonup x^\star \in \operatorname{Fix} T. \tag{3.3}$$

For $s = 1$, the coefficient of $x_k - x_{k-1}$ in (3.2) vanishes, and the iteration becomes

$$x_{k+1} = \left(1 - \frac{\alpha}{2(k+\alpha)}\right) x_k + \frac{\alpha}{2(k+\alpha)} Tx_k + \frac{k}{k+\alpha}(Tx_k - Tx_{k-1}). \tag{3.4}$$

As observed in [1, Remark 2.1], this is a KM iteration with the additional extrapolation term $\frac{k}{k+\alpha}(Tx_k - Tx_{k-1})$. The extrapolation acts on the operator values. For $0 < s < 1$, the explicit inertial term $\frac{(1-s)k}{k+\alpha}(x_k - x_{k-1})$ is also present.

3.3 Stochastic Fast KM

We replace the exact evaluations in (3.2) by the conditionally unbiased oracle responses from Section 2.2. Choose deterministic $x_0, x_1 \in \mathcal{H}$ and fixed $\alpha > 2$, $0 < s \leq 1$. Initialize by computing and storing $\widehat{T}_0(x_0)$. For every $k \geq 1$, compute $\widehat{T}_k(x_k)$ and set

$$\begin{aligned} x_{k+1} = &\left(1 - \frac{s\alpha}{2(k+\alpha)}\right) x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)} \widehat{T}_k(x_k) \\ &+ \frac{sk}{k+\alpha}\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big). \end{aligned} \tag{3.5}$$

The preceding response $\widehat{T}_{k-1}(x_{k-1})$ is stored from the previous iteration. Hence, after initialization, each update requires one new oracle evaluation. The points x_{k-1} and x_k, together with the stored response, are $\mathcal{F}_k$-measurable; x_{k+1} is $\mathcal{F}_{k+1}$-measurable. The parameters are independent of a terminal horizon.

Substituting (2.5) into the difference of oracle responses yields

$$\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1}) = Tx_k - Tx_{k-1} + e_k - e_{k-1}. \tag{3.6}$$

Relative to the exact update (3.2) evaluated at the current iterates, the total additive error is

$$\eta_k = \frac{s\alpha}{2(k+\alpha)} e_k + \frac{sk}{k+\alpha}(e_k - e_{k-1}) = \frac{s(k+\alpha/2)}{k+\alpha} e_k - \frac{sk}{k+\alpha} e_{k-1}. \tag{3.7}$$

Since e_{k-1} is $\mathcal{F}_k$-measurable, conditional unbiasedness gives

$$\mathbb{E}[\eta_k \mid \mathcal{F}_k] = -\frac{sk}{k+\alpha} e_{k-1}. \tag{3.8}$$

These identities describe how the current and preceding oracle errors enter the stochastic iteration.

Our objective is to identify noise conditions under which

$$k\|x_k - Tx_k\| \longrightarrow 0 \qquad \text{almost surely,} \tag{3.9}$$

and, for a $\operatorname{Fix} T$-valued random variable x_∞,

$$x_k \rightharpoonup x_\infty \qquad \text{almost surely.} \tag{3.10}$$

References

[1] Radu Ioan Boţ and Dang-Khoa Nguyen. Fast Krasnosel'skiĭ–Mann algorithm with a convergence rate of the fixed point iteration of $o(1/k)$. *SIAM Journal on Numerical Analysis*, 61(6):2813–2843, November 2023. ISSN 1095-7170. doi: 10.1137/22m1504305. URL http://dx.doi.org/10.1137/22M1504305.

[2] Mario Bravo and Roberto Cominetti. Stochastic fixed-point iterations for nonexpansive maps: Convergence and error bounds. *SIAM Journal on Control and Optimization*, 62(1):191–219, January 2024. ISSN 1095-7138. doi: 10.1137/22m1515550. URL http://dx.doi.org/10.1137/22M1515550.

[3] Nghia Nguyen-Trung and Quoc Tran-Dinh. A class of accelerated fixed-point-based methods with delayed inexact oracles and its applications, 2025. URL https://arxiv.org/abs/2512.13547.

[4] Tong Yang, Tao Jiang, Yuejie Chi, Ashok Cutkosky, and Lin Xiao. Stochastic inertial krasnosel'skii-mann iteration achieves near-optimal sample complexity, 2026. URL https://arxiv.org/abs/2609.28543.